

CoRoT-2b

R-2090

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_110504 · created 2026-08-10T00:51:33+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455685.9625 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.064
Transit depth (Rp/Rs)^2	4e-05 +/- 0.0081 [%]
Orbital Inclination (inc)	85.0 +/- 2.05
Airmass coefficient 1 (a1)	0.93 +/- 0.3
Airmass coefficient 2 (a2)	0.06 +/- 0.4
Scatter in the residuals of the lightcurve fit is	32.89 %
Best Comparison Star	#1 - [487, 163]
Optimal Aperture	5.93
Optimal Annulus	14.09
Transit Duration (day)	0.068 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.001 $\pm$ 0.064 vs 0.1667 (deviation 2.55 σ)
Duration fit vs published	0.068 d vs 0.095 d (deviation 1.56 σ)
Mid-time O-C	-6.4 min
Depth SNR	0.0
Residual scatter	32.89 %

Light curve

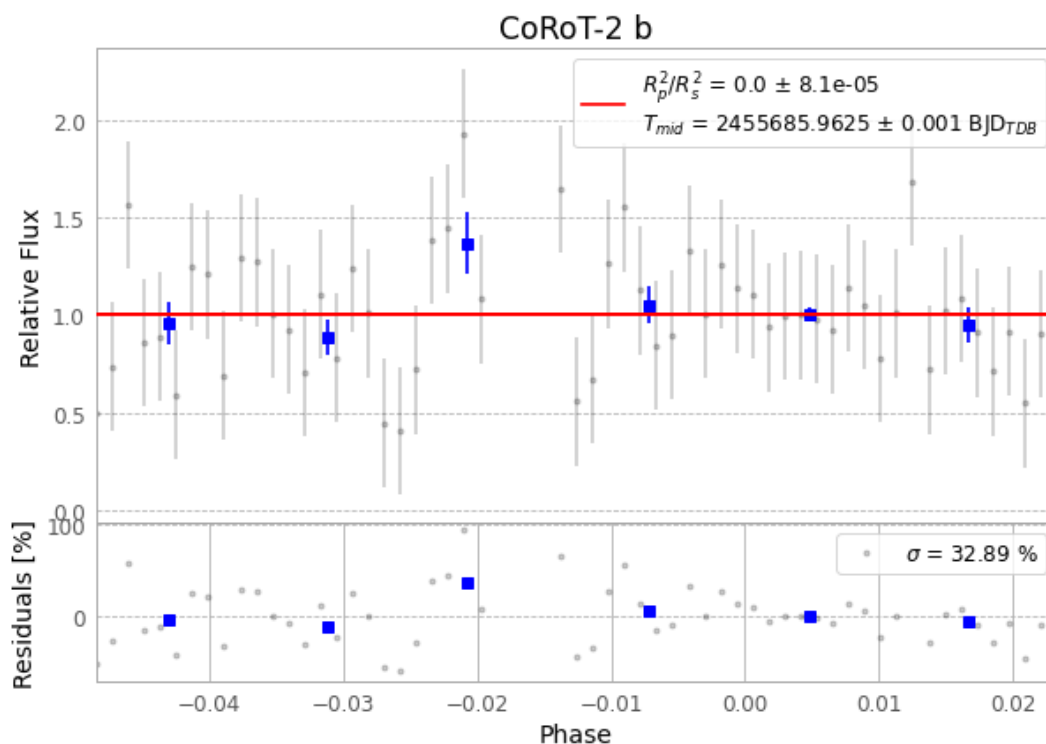


Figure 1 — FinalLightCurve_CoRoT-2 b_2011-05-04.png (SHA-256 f4a5a1c0387b2087...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [284, 399], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 570185e4a4811347...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

61 FITS frames (40 MB), CoRoT-2110504090012.FITS → CoRoT-2110504120013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-110504.log (44359fae183c...), FinalLightCurve_CoRoT-2 b_2011-05-04.png (f4a5a1c0387b...), FinalLightCurve_CoRoT-2 b_2011-05-04.csv (305eea94a76e...)

Compute resources

Wall time 162.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 00:51Z.